

SHEKHAR PRATYUSH

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PROFESSIONAL SUMMARY

Motivated 2nd-year B.Tech CSE student specializing in AI/ML at Bennett University, with hands-on experience in Machine Learning, Deep Learning, Computer Vision, and full-stack AI application development. AWS Certified AI Practitioner with a strong foundation in building real-world intelligent systems — from AI interview trainers to indoor navigation apps for the visually impaired. Passionate about solving impactful problems through data-driven approaches and cutting-edge AI techniques.

AREAS OF INTEREST

Machine Learning | Deep Learning | Computer Vision | Natural Language Processing | Competitive Programming | Full-Stack AI Development | Predictive Modeling

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL, JavaScript, React, Node.js, Typescript

ML / DL Libraries: Scikit-learn, TensorFlow, PyTorch, OpenCV, NLTK, NumPy, Pandas, Matplotlib, Seaborn

Machine Learning: Regression, Classification, Clustering, Ensemble Methods, TF-IDF, Cosine Similarity

Deep Learning: Neural Networks, CNN, RNN, LSTM, Computer Vision pipelines

Tools & Frameworks: FastAPI, Streamlit, Supabase, Docker, Jupyter Notebook, Google Colab

Cloud & IDEs: AWS, Jupyter Notebook, Google Colab, VS Code, Git/GitHub

PROJECTS

Persona.ai — AI Interview Trainer

React · Node.js · Groq AI · UE5 MetaHuman · Python

- Built a full-stack AI-powered mock interview platform featuring a UE5.7 MetaHuman avatar (Payton) with real-time lip sync via Rhubarb and Edge TTS.
- Engineered multi-turn interview logic with intro phase detection, cross-questioning, and a sequential /answer → /react → /next conversation flow using Groq AI.
- Deployed frontend to Vercel and backend to Railway via Docker, integrating Supabase as the database with connection pooler URLs for production reliability.
- Implemented Pixel Streaming to render the 3D avatar in-browser with minimal latency, delivering an immersive interview simulation experience.

BlindNav — Indoor Navigation for the Visually Impaired

TypeScript · TSX · TFLite · WiFi Fingerprinting

- Developed a hackathon project (Hackaccino) enabling indoor navigation for blind students using WiFi fingerprinting and a KNN classifier (scikit-learn) on a Pixel 7a.
- Integrated COCO SSD TFLite model for real-time obstacle detection with a 40% confidence threshold, bridged to React Native state via Worklets.createRunOnJS.
- Implemented Text-to-Speech (expo-speech) with a 4-second throttle to deliver clear, non-overwhelming audio navigation cues.

Movie Recommendation System

Python · Scikit-learn · TF-IDF · Streamlit

- Built a content-based movie recommendation engine using TF-IDF vectorization and cosine similarity to match user preferences with relevant titles.
- Processed and cleaned movie metadata to extract meaningful features, improving recommendation relevance across diverse genres.
- Implemented and tested entirely within Jupyter Notebook, with results and recommendations displayed inline using code-block outputs.

HateMM — Vision-Language Classification for Social Media Moderation

Python · PyTorch · ViT-B/16 · RoBERTa · MMHS150K

- Built a 3-modality hate speech classifier (image + text caption + OCR text) on the MMHS150K dataset (149,823 real social media posts) as an AI/ML academic project at Bennett University.
- Designed a Cross-Modal Transformer (6 layers, 8 heads) with a custom Modality-Gated Attention (MGA) mechanism that learns per-sample scalar weights for each modality, enabling interpretable and adaptive fusion.
- Diagnosed and resolved multimodal collapse — V1 had 74% text gate dominance (AUROC 0.748); fixed by freezing RoBERTa to force ViT to learn visual features, achieving balanced gate weights (0.33/0.33/0.33).
- Implemented a 3-phase training pipeline: contrastive pretraining → full fine-tuning with early stopping → hard negative mining, achieving 75.96% accuracy and AUROC 0.714 on the test set.

LeetCode Heatmap Widget

Java · Android · GraphQL API

- Developed an Android home-screen widget that fetches and displays a user's LeetCode submission heatmap using the GraphQL API.
- Implemented efficient background data refresh to keep contribution stats up-to-date without draining battery.

EDUCATION

Bachelor of Technology in Computer Science Engineering — AI/ML Specialization

2024 – 2028

Bennett University, Greater Noida

Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems

CERTIFICATIONS

- **AWS Certified AI Practitioner** — Amazon Web Services (AWS)
- **Computer Networks** — Coursera
- **Operating Systems** — Coursera
- **Design Thinking & Innovation (DTI)** — Coursera